# ECON 2 Practice final A

Name: _____ SID: _____ · 90 minutes · 60 points · one page of notes, front and back · calculator allowed

## The exam

Clear your desk except for your cheat sheet, a calculator, a pen, a ruler, and blank paper. For each graph part, copy the printed axes onto your paper with the same tick marks, because on exam day the axes are already printed and your job is to draw on them. Start a 90-minute timer and do not stop it.

The exam is worth 60 points in 90 minutes, so a point is worth about a minute and a half. A sensible budget is about 18 minutes for Question 1, 15 minutes for Question 2, 36 minutes for Question 3, and 18 minutes for Question 4. His cover page says "Less is more." A 2-point answer is the right answer plus one or two sentences of reason in course words.

After time is called, grade one part at a time. Open "Show answer," compare your work to the model answer and the "Full credit" line, and give yourself full, half, or zero credit honestly. Then use the table in "Score it" to decide what to reread.

### Free points (2 points)

**1 point** Write your name and student ID at the top of each page of the exam.

**1 point** If the Sorting Hat at Hogwarts put you in a house, which house would it be? Always remember: the Hat takes your own choice into account. (There are no wrong answers.)

### Question 1 (12 points total): Circle back to the whole semester

For each of these questions, circle the one correct answer. Each part is worth 2 points.

**2 points** 1(a). A wheat farmer who sells in a perfectly competitive market maximizes profit by producing the quantity at which

(i) Price equals average total cost
(ii) Total revenue is as large as possible
(iii) Marginal revenue equals marginal cost
(iv) Marginal cost is as low as possible

**2 points** 1(b). A cement plant releases dust that harms the health of families living nearby, and nobody makes the plant pay for that harm. Compared with the efficient quantity, the unregulated market for cement produces

(i) Too much cement, because social marginal cost is above private marginal cost
(ii) Too little cement, because social marginal benefit is above private marginal benefit
(iii) The efficient quantity, because the market sets marginal benefit equal to private marginal cost
(iv) Too little cement, because the plant's profits are too low

**2 points** 1(c). A Berkeley professor can grade exams a little faster than any of her graduate student instructors (GSIs), and she can write research papers much faster than any of them. The principle of comparative advantage implies that

(i) She should grade the exams herself, because she is the fastest grader
(ii) She should do both tasks herself, because she is faster at both
(iii) The GSIs should write the research papers, because that task is respected most
(iv) She should specialize in research and let the GSIs grade, because her opportunity cost of grading is higher than theirs

**2 points** 1(d). Your savings account pays a nominal interest rate of 5% per year, and the inflation rate is 3% per year. Your real interest rate is

(i) 8%
(ii) 2%
(iii) 15%
(iv) −2%

**2 points** 1(e). The economy is in a recession, with output below potential output. To fight the recession, the Federal Reserve would typically

(i) Raise the real interest rate to make saving more attractive
(ii) Raise government purchases G to shift planned aggregate expenditure up
(iii) Lower the real interest rate, which raises interest-sensitive spending such as investment and new car purchases
(iv) Do nothing, because prices and wages adjust right away

**2 points** 1(f). Which of these counts in this year's U.S. gross domestic product (GDP)?

(i) Your neighbor buys your used 2019 car
(ii) You buy shares of stock in a technology company
(iii) A bakery buys flour that it uses to bake the bread it sells
(iv) A household buys a new air conditioner that was built in Ohio this year

### Question 2 (10 points total): Short answers, long-run payoffs

Each part is worth 2 points. Show your work where there is arithmetic, and briefly explain.

**2 points** 2(a). Suppose a movie ticket cost $8.00 in 2000 and $20.00 in 2025, and suppose the consumer price index (CPI, with 1982 to 1984 = 100) was 160 in 2000 and 320 in 2025. These are round numbers made up for this question. Did the price of a movie ticket rise, fall, or stay the same in inflation-adjusted (real) terms? That is, if your income rose at the rate of inflation, could you buy more, fewer, or the same number of tickets in 2025 as in 2000? Show your work.

**2 points** 2(b). A bond promises to pay $630 one year from now and $661.50 two years from now, and nothing else. If the interest rate is 5% per year, what is the most a buyer would pay for this bond today? If the Fed then raises interest rates and the market interest rate on this kind of bond rises to 8%, does the price of the bond rise or fall? Show your work.

**2 points** 2(c). Two countries start with the same real GDP per person, $10,000. In Country A it grows 2% per year, and in Country B it grows 1% per year. Use the rule of 70 to find about how many years it takes each country's real GDP per person to double. About how much is real GDP per person in each country after 70 years?

**2 points** 2(d). Suppose the Fed raises U.S. interest rates and the exchange rate moves from €0.80 per U.S. dollar to €0.90 per U.S. dollar. Has the dollar appreciated or depreciated? What happens to the price in euros of a $50 pair of jeans made in the U.S., and what happens to U.S. net exports?

**2 points** 2(e). A small town has 4,000 adults. Of them, 2,280 have jobs, 120 have no job but have actively looked for work in the past four weeks, and the other 1,600 are retired, in school full time, or not looking for work. What is the town's unemployment rate? Now suppose 25 of the job seekers become discouraged and stop looking. What is the new unemployment rate, and is this good news for the town? Show your work.

## Question 3 (24 points total): Data centers go dark, and we Keynesian Cross our fingers

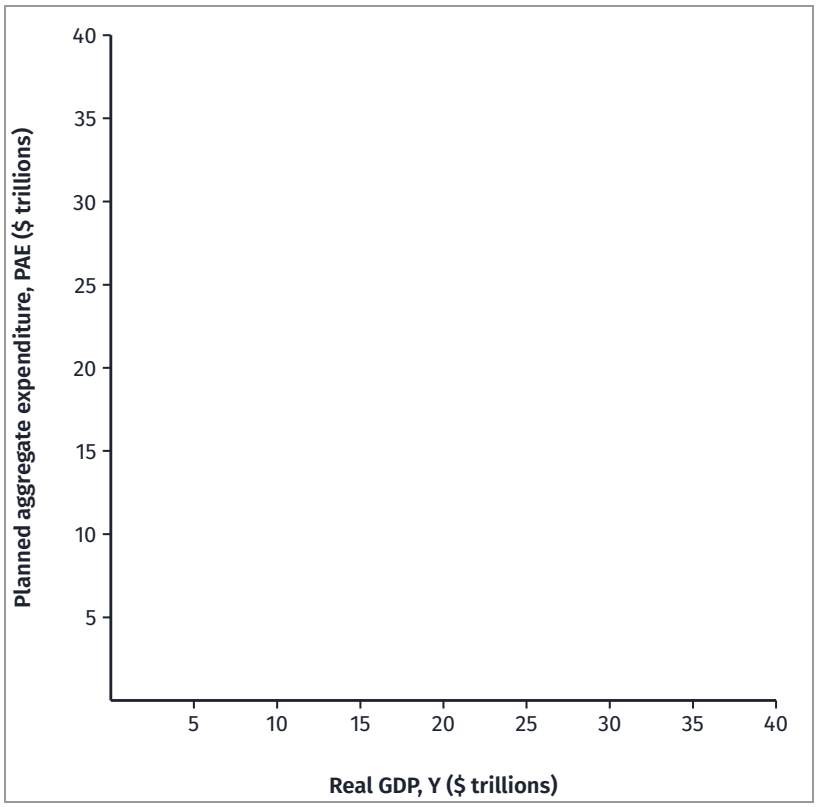
Investment in AI data centers, the huge buildings full of computer chips and power equipment that train and run AI models, has helped fuel recent growth in U.S. GDP. Suppose U.S. GDP is about $30 trillion, and suppose the aggregate marginal propensity to consume out of disposable income is mpc = ½. The following system models planned aggregate expenditure (PAE) in trillions of dollars, where T is net taxes:

| | | | |
|---|---|---|---|
| **Consumption** | $C = 8 + \frac{1}{2}(Y - T)$ | **PAE** | $PAE = C + I + G + NX$ |
| **Net taxes** | $T = 6$ | **Investment** | $I = 6$ |
| **Government purchases** | $G = 6$ | **Net exports** | $NX = -2$ |

**2 points** 3(a). Combine the expenditure terms in PAE and find the level of PAE when Y = 0, which is the vertical intercept in the Keynesian cross diagram. Show your work.

**2 points** 3(b). Using your answer to part (a), set PAE = Y and solve for the short-run equilibrium level of Y, which is the horizontal coordinate of the spot where PAE crosses the 45° line. Show your work.

**4 points** 3(c). Graph the PAE line and the 45-degree line on the axes below, and show the equilibrium level of output in the short run. Label intercepts, slopes, and the equilibrium.




Printed axes for 3(c). Both axes use the same scale. Copy them onto your paper.

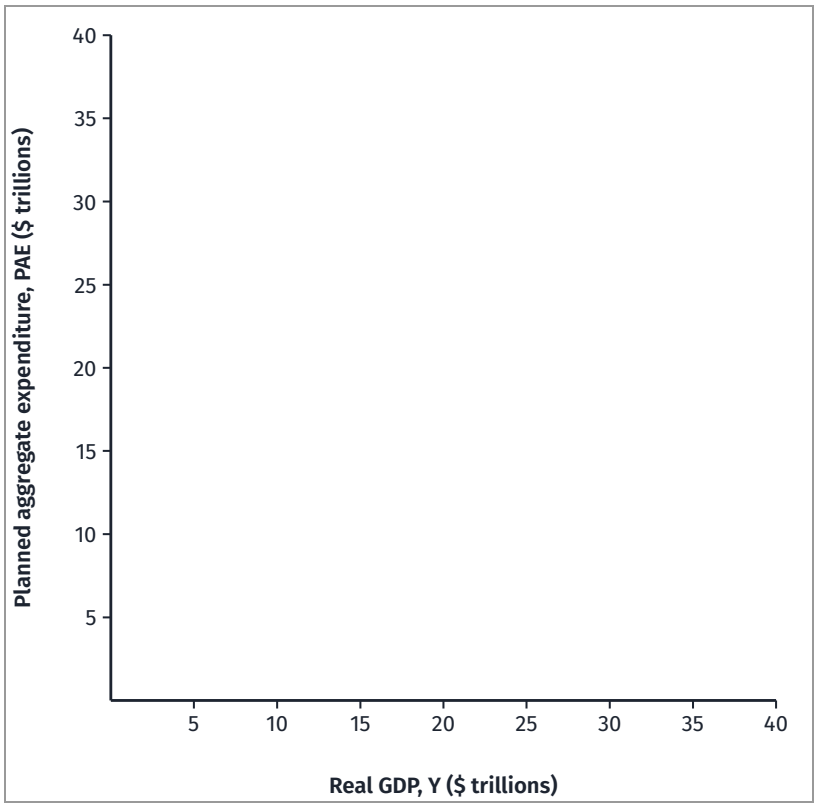
In 2027, suppose several large technology companies announce that they will pause new data-center projects, because they now expect AI services to earn less in the future than they had hoped. Stock prices of AI-related companies fall, and households who own those stocks feel less wealthy. Suppose these events produce this situation:

| | | | |
|---|---|---|---|
| **Consumption** | C' = 7.5 + ½(Y − T) | **PAE'** | PAE' = C' + I' + G + NX |
| **Net taxes** | T = 6 | **Investment** | I' = 4.5 |
| **Government purchases** | G = 6 | **Net exports** | NX = −2 |

**2 points** 3(d). Combine the expenditure terms in PAE' and find the level of PAE' when Y = 0, the new vertical intercept. Show your work.

**2 points** 3(e). Using your answer to part (d), set PAE' = Y and solve for the new short-run equilibrium level of Y. Show your work.

**4 points** 3(f). On the axes below, graph the original PAE line from part (a), the new PAE' line from part (d), and the 45-degree line. Show both equilibria and label intercepts.




Printed axes for 3(f). Copy them onto your paper.

**2 points** 3(g). By how much did planned aggregate expenditure change between (a) and (d)? Express your answer in trillions of dollars and describe it as an increase, a decrease, or no change. Explain fully, saying which spending components changed and why, and describe how the change appears in the graph.

**2 points** 3(h). By how much did GDP change between (b) and (e)? Is that change the same size as your answer to (g), or different? If different, say which is larger and explain why, referencing the model.

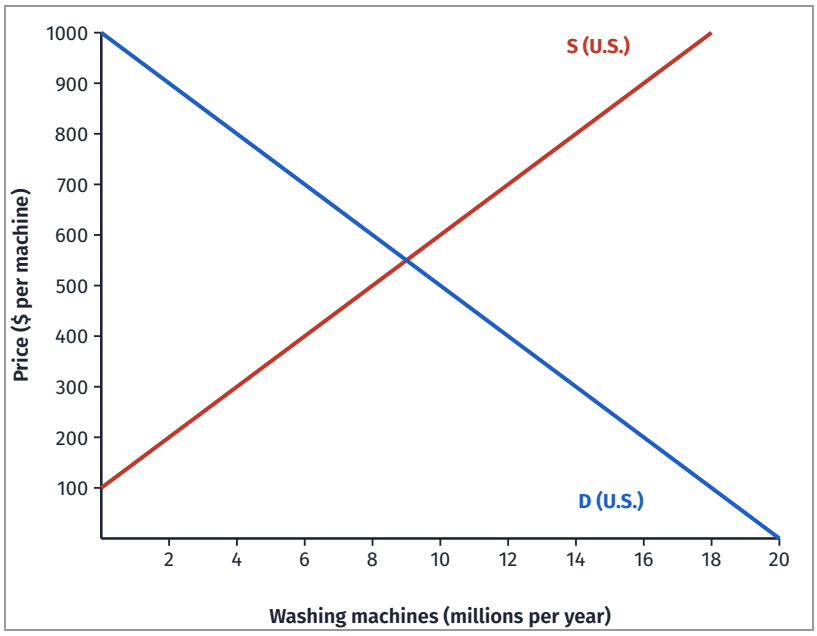
**2 points** 3(i). Suppose Congress could quickly pass a temporary spending bill that changes government purchases G by enough to bring GDP back to $30 trillion. How big a change in G is that? Is it an increase or a decrease? Explain fully.

**2 points** 3(j). Instead of raising G, suppose Congress cuts net taxes T. How big a tax cut would bring GDP back to $30 trillion? Is it larger or smaller than your answer to (i)? Explain fully, and say which policy adds more to the government budget deficit.

### Question 4 (12 points total): Spin cycle: washing machines, tariffs, and jobs

Washing machines are a familiar tariff story in this class. Suppose the U.S. market for washing machines has domestic demand $P = 1000 - 50Q$ and domestic supply $P = 100 + 50Q$, where P is dollars per machine and Q is millions of machines per year. The U.S. is a price taker on the world market: it can buy as many machines as it wants at the world price of $300.

**4 points** 4(a). The diagram below shows domestic supply and demand. Find the price and quantity if the U.S. did not trade. Then adapt the diagram to show free trade at the world price of $300: show the world price, the domestic quantity supplied, the domestic quantity demanded, and the quantity of imports, with values. Briefly explain what you drew.



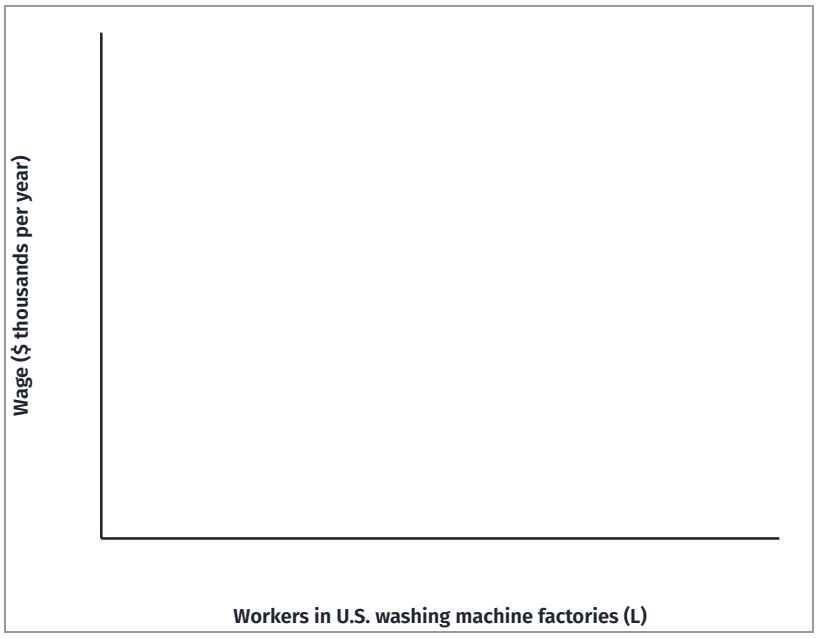

Printed diagram for 4(a) and 4(b): U.S. supply and demand for washing machines. Copy it onto your paper.

**2 points** 4(b). Now the U.S. puts a tariff of $100 per machine on imported washing machines. What is the new price in the U.S.? What are the new domestic quantity supplied, domestic quantity demanded, and imports? Show the tariff on your graph from (a).

**2 points** 4(c). Calculate the tariff revenue the government collects and the total deadweight loss of the tariff. Shade and label the deadweight loss areas on your graph, and briefly explain why each one is a loss.

Now consider the labor market for workers in U.S. washing machine factories. There is a labor supply curve, reflecting workers' marginal cost of supplying their labor, and a labor demand curve equal to the marginal revenue product of labor, MRPL = MR × MPL, where MPL is the marginal product of labor (the extra machines one more worker produces) and MR is the marginal revenue from selling one more machine.

**4 points** 4(d). U.S. washing machine makers are price takers, so their marginal revenue is the U.S. price of a machine. Before the tariff, the last worker hired adds 150 machines per year (MPL = 150). What is that worker's MRPL before the tariff, and what is it after the tariff, at the same level of employment? Illustrate the labor market before and after the tariff, showing all demand and supply curves and the equilibrium wages and employment. Does the tariff raise total U.S. employment? Briefly explain.




Printed axes for 4(d). Copy them onto your paper.